

# YUXIN YANG

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## RESEARCH INTERESTS

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Large Language Models, Agentic AI Systems, Generative Recommendation, Retrieval-Augmented Generation, Test-time Scaling, Graph Machine Learning

## EDUCATION

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**Ph.D. in Computer Science**, University of Southern California

Sep 2023 – present

- **GPA:** 3.83/4.00
- **Core courses:** Parallel and Distributed Computation (4.0), Advanced Analysis of Algorithms (4.0), Theoretical Machine Learning (4.0), The Science of Large Language Models (4.0)

**B.Eng. in Automation**, Tsinghua University

Sep 2019 – Jun 2023

- **GPA:** 3.84/4.00

## SELECTED AWARDS AND HONORS

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- Capital One PhD Fellow 2024–2025, 2025–2026, 2026–2027
- Changtong Scholarship (top 1% in the Department) 2022
- Comprehensive Scholarship of University (top 5% in University) 2020, 2021

## COMPUTER AND LANGUAGE SKILLS

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- **Programming:** Python, C/C++, JavaScript
- **Tools:** PyTorch, MATLAB, CMake, LabVIEW, Linux, Git, LaTeX
- **Language:** TOEFL 114/120 (Reading 30, Listening 29, Speaking 25, Writing 30); GRE 327/340+3.5/6.0 (Verbal 157, Quantitative 170, Analytical Writing 3.5)

## PUBLICATIONS AND MANUSCRIPTS

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**Yuxin Yang**, Gangda Deng, Ömer Faruk Akgül, Nima Chitsazan, Yash Govilkar, Akasha Tigalappanavara, Shi-Xiong Zhang, Sambit Sahu, Viktor Prasanna. “SPARC-RAG: Adaptive Sequential-Parallel Scaling with Context Management for Retrieval-Augmented Generation.” Conference on Empirical Methods in Natural Language Processing (EMNLP Main Conference), 2026 (accepted).

Gangda Deng, Zhaoling Chen, Zhongming Yu, Haoyang Fan, Yuhong Liu, **Yuxin Yang**, Dhruv Parikh, Rajgopal Kannan, Le Cong, Mengdi Wang, Qian Zhang, Viktor Prasanna, Xiangru Tang, Xingyao Wang. “SWE-Milestone: Evaluating AI Agents on Continuous Software Evolution.” International Conference on Machine Learning (ICML), 2026.

Abdulla Alshabanah, **Yuxin Yang**, Murali Annavaram. “SAGERec: Sampling and Gating for Enhanced Long-Tail Item Recommendations.” ACM International Conference on Web Search and Data Mining (WSDM), 2026.

Gangda Deng\*, **Yuxin Yang\***, Ömer Faruk Akgül\*, Hanqing Zeng, Yinglong Xia, Rajgopal Kannan, Viktor Prasanna. “Training Diverse Graph Experts for Ensembles: A Systematic Empirical Study.” Preprint, 2025.

Ömer Faruk Akgül\*, Feiyu Zhu\*, **Yuxin Yang**, Rajgopal Kannan, Viktor Prasanna. “RECIPE-TKG: From Sparse History to Structured Reasoning for LLM-based Temporal Knowledge Graph Completion.” Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2026.

**Yuxin Yang**, Hongkuan Zhou, Rajgopal Kannan, Viktor Prasanna. “Towards Ideal Temporal Graph Neural Networks: Evaluations and Conclusions after 10,000 GPU Hours.” 51st International Conference on Very Large Data Bases (VLDB), 2025.

**Yuxin Yang**, Yitao Liang, and Muhan Zhang. “PA-GNN: Parameter-Adaptive Graph Neural Networks.” Workshop on Dynamic Neural Networks in *the 39th International Conference on Machine Learning (ICML-DyNN-Workshop)*, 2022, oral presentation.

## INDUSTRY EXPERIENCE

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### ByteDance

May – Aug 2026

*Summer Intern, LLM-native Generative Recommendation*

Used semantic ID (SID) sequences from users' historical item interactions as LLM inputs to directly predict the SID of the next interacted item.

- **Training and evaluation:** Built the training and offline evaluation environment from scratch, including a daily fine-tuning framework. Designed two-stage pretraining + SFT experiments; implemented **next- $N$  prediction** and its training kernel, using one shared history context to predict the next  $N$  item interactions.
- **Modeling studies:** Studied model-size and context-length scaling, observing modest positive gains from larger models. Designed **SID-text alignment** experiments: alignment-based initialization followed by recommendation pretraining outperformed starting directly from pretrained Qwen weights.
- **Serving acceleration:** Designed a **tree-decoding** acceleration scheme and contributed to **BF16-to-FP8** conversion and validation, observing evaluation metric differences below **0.0002**.
- **Online results:** Subsequent online experiments showed gains of approximately **1.5× the internal shipping bar**. This does not imply launch-review approval.

## RESEARCH EXPERIENCE

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### SPARC-RAG: Adaptive Scaling and Context Management for RAG

Jan 2025 – Jan 2026

Research Assistant, advisor: Prof. Viktor Prasanna

University of Southern California

*In collaboration with Capital One Research*

- Proposed **SPARC-RAG**, a multi-agent framework that adaptively scales **sequential depth** and **parallel width** for multi-hop Retrieval Augmented Generation (RAG), addressing context contamination and scaling inefficiency in naive scaling approaches.
- Designed a **unified context management** mechanism with specialized agents maintaining shared global context, enabling explicit control over the scaling process and avoiding context pollution.
- Implemented **dynamic query generation** that produces targeted, complementary sub-queries for each branch, enabling diverse parallel exploration of the reasoning space.
- Developed **lightweight fine-tuning** with process-level verifiable preferences for adaptive exit decisions based on answer correctness and evidence grounding.
- Achieved **+6.2 average F1 improvement** on multi-hop QA benchmarks (HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle) under lower inference cost.

### Towards Ideal Temporal Graph Neural Networks

Oct 2023 – May 2024

Research Assistant, advisor: Prof. Viktor Prasanna

University of Southern California

- Designed a unified and fully optimized TGNN comparison framework that isolates and evaluates core modules (neighbor sampling, node memory, aggregation), enabling fair design-space exploration across models.
- Conducted a detailed experimental study to answer three key research questions on efficiency, universality, and module interplay; identified that most-recent sampling, attention aggregation, and node memory yield optimal cost-performance tradeoffs.
- Discovered that dataset repetition patterns fundamentally determine node-memory preference, showing RNN-based memory excels on short-term repetition datasets while static embedding memory excels on long-term repetition datasets.
- Revealed strong diminishing returns in deeper/wider neighbor sampling when node memory is present, leading to practical design recommendations for future TGNN architectures.

### PA-GNN: Parameter-adaptive Graph Neural Networks

Nov 2021 – Jun 2022

Research Assistant, advisor: Prof. Muhan Zhang

Beijing Institute for General Artificial Intelligence

- Showed that capturing local patterns of graph datasets improves performance of learning tasks on graphs.
- Designed a trainable node-specific aggregator that learns from node position and features to capture local patterns; Utilized DeepWalk to encode node position.
- Experimentally showed that graph neural networks with an adaptive aggregator can exploit local patterns and improve node classification accuracy by up to 5%.